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# =====
# 18. Feature Detection in Noisy Images
# Google Colab Implementation using ORB
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# STEP 1: Upload Image
# -----
print("Upload an image (jpg/png)")
uploaded = files.upload()

img_path = next(iter(uploaded))

# -----
# STEP 2: Read Image
# -----
img = cv2.imread(img_path)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -----
# STEP 3: Add Gaussian Noise
# -----
noise = np.random.normal(0, 25, gray.shape).astype(np.uint8)
noisy = cv2.add(gray, noise)

# -----
# STEP 4: Create ORB Feature Detector
# -----
orb = cv2.ORB_create(nfeatures=300)

# Detect features on original image
kp_original, des1 = orb.detectAndCompute(gray, None)

# Detect features on noisy image

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kp_noisy, des2 = orb.detectAndCompute(noisy, None)

# -----
# STEP 5: Draw Keypoints
# -----
original_features = cv2.drawKeypoints(
    img_rgb, kp_original, None, color=(0,255,0), flags=0)

noisy_features = cv2.drawKeypoints(
    noisy, kp_noisy, None, color=(255,0,0), flags=0)

# -----
# STEP 6: Display Results
# -----
plt.figure(figsize=(15,8))

# Original image
plt.subplot(2,2,1)
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

# Noisy image
plt.subplot(2,2,2)
plt.imshow(noisy, cmap='gray')
plt.title('Noisy Image')
plt.axis('off')

# Features on original image
plt.subplot(2,2,3)
plt.imshow(original_features)
plt.title(f'Original Features: {len(kp_original)}')
plt.axis('off')

# Features on noisy image
plt.subplot(2,2,4)
plt.imshow(noisy_features, cmap='gray')
plt.title(f'Noisy Features: {len(kp_noisy)}')
plt.axis('off')

plt.suptitle('Feature Detection in Noisy Images using ORB',
    fontsize=18, fontweight='bold')

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plt.tight_layout()
plt.show()

# -----
# STEP 7: Print Feature Counts
# -----
print("Number of features in Original Image :", len(kp_original))
print("Number of features in Noisy Image      :", len(kp_noisy))

```

Upload an image (jpg/png)

img1.jpeg

img1.jpeg(image/jpeg) - 14202 bytes, last modified: 20/07/2026 - 100% done

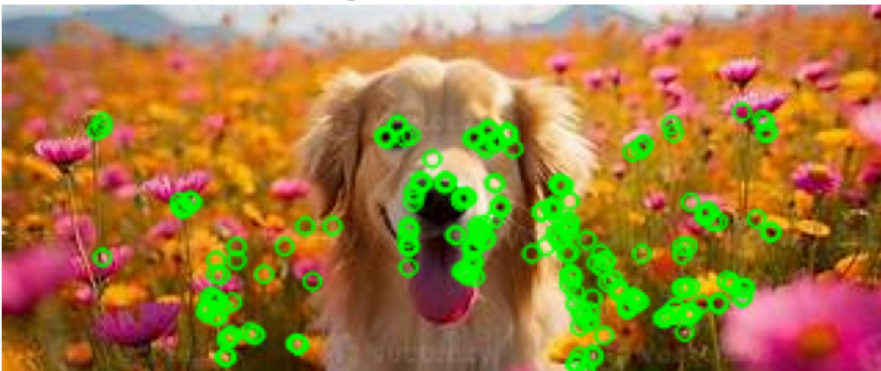
Saving img1.jpeg to img1.jpeg

Feature Detection in Noisy Images using ORB

Original Image



Original Features: 254



Noisy Image



Noisy Features: 259

